

THE STARRY MESSENGER

The Global Society of Young Physicists | Issue 3 | April 2025



“So what really is Dark Matter?” by Sherlyn Nammi

“Dark Energy, Black Holes and the End of the Universe: Is there hope for civilisation at the end?” by Jason Pettigrew

“RMP 2024 Reflections” by Jana Bazeed

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We are thrilled to bring you the third edition of *The Starry Messenger*, the triannual science magazine of the Global Society of Young Physicists (GSYP). This issue introduces the extraordinary work of physicists worldwide, featuring two remarkable research articles and an interview article on:

- **Dark Energy, Black Holes and the End of the Universe: Is there hope for civilisation at the end?**
- **So what really is Dark Matter?**
- **RMP 2024 Reflections**

The GSYP was founded with the mission of uniting young physics enthusiasts across the globe to foster a vibrant community dedicated to advancing physics education. As a registered community interest company, we provide budding physicists with various opportunities, including the *Research Mentorship*

Programme (RMP), where students collaborate with experienced mentors on challenging research projects.

We warmly invite you to support and join us in our mission to promote physics education and connect young enthusiasts worldwide. If you are interested in contributing to our science magazine as an author, we encourage you to submit your work, we want to hear your voice! Details can be found on the posters included at the end of this issue.

This issue is dedicated to Aalia Mahomed, who was a Physics student at King's College London. You were the very first person I met on our course, Al, and I will treasure your memory forever.

Arya Lal Gonullu

Chief Executive

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This issue of The Starry Messenger is dedicated to Aalia Mahomed, whose loss is felt throughout our community. Aalia was a bright, young physics student at King's College London, and we are deeply saddened by the loss of her.

- The Starry Messenger team

Become a Member of GSYF!

As a member of GSYF, you gain access to exclusive benefits that will enhance your journey as a young physicist. Members can request reviews from research mentors who are conducting cutting-edge studies in Physics. After a thorough editorial review, you can submit your work for publication in our science magazine. Additionally, you will have the opportunity to attend talks by renowned science experts brought in by GSYF.

Membership is currently free, so join us and take advantage of these incredible opportunities.

Science Magazine Schedule

Our Science Magazine is published three times a year:

- ❖ **April:** Independent research published before the annual RMP.
- ❖ **August:** Research reports from the annual RMP.
- ❖ **December:** Independent research published after the annual RMP.

Stay tuned for our next issue in April 2025, where we will feature more research pieces and provide details on the upcoming RMP 2025!

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We invite you to become a GSYF member and take the first step towards your research dreams. Access expert talks, publish your research, and connect with a global community of young physicists. Don't miss out on this incredible opportunity to unlock your potential as a physics researcher.

*The views of our writers are their own.
We aim for sound science but not editorial orthodoxy.*

So what really is Dark Matter?

Author: Sherlyn Nammi

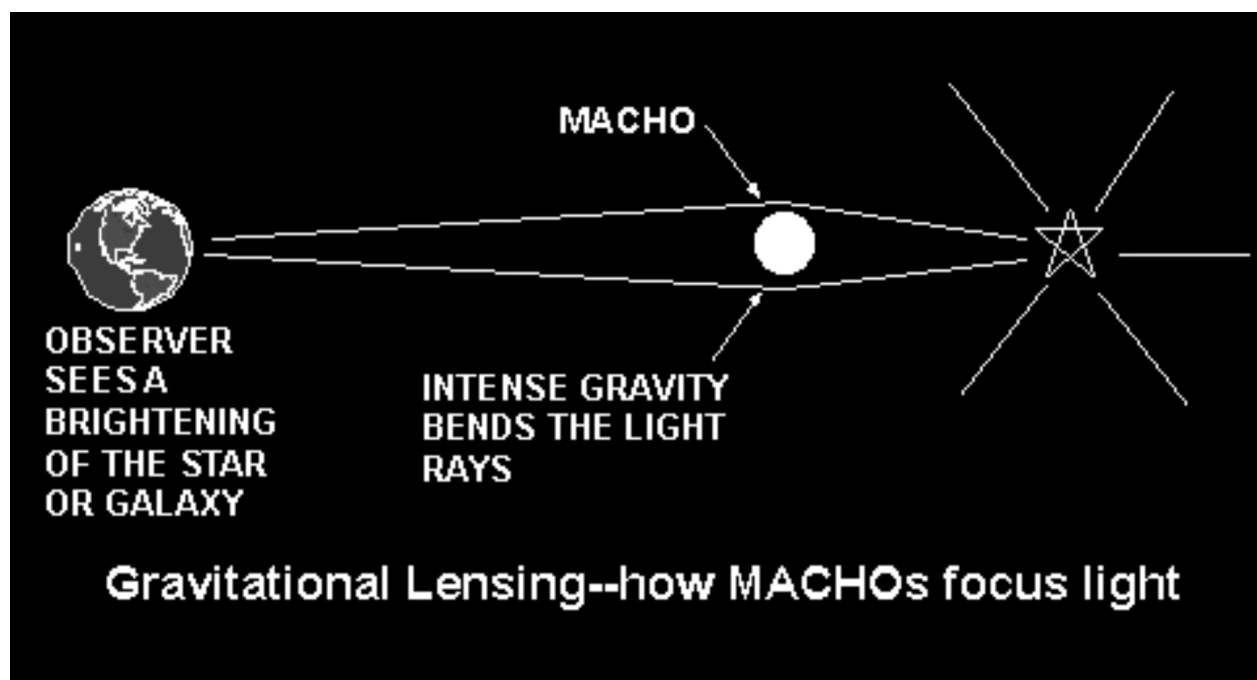
As modern technology has advanced, our scientific understanding of the tendencies and movements of celestial bodies have been applied and verified observationally, furthering our models of understanding of our universe [1]. But gradually, physicists began to notice that something in their calculations wasn't quite adding up.

In this article, Sherlyn Nammi, a student of Physics with Astrophysics & Cosmology at King's College London, explores the current theories on the nature of dark matter in the universe.

In 1933, Fritz Zwicky of Caltech university was using the mount Wilson observatory to study a cluster of galaxies. He published his historic paper titled "Die Rotverschiebung von extragalaktischen Nebeln" -the redshift of extragalactic nebulae. He was the first person to notice something rather peculiar about the movement of these galaxies. He stated: "in order to obtain the observed value of (velocity), the average density in the Coma system would have to be at least 400 times larger than that derived on the grounds of observations of luminous matter". This suggestion led to the first ever conclusion that in order to explain this phenomenon and make it abide by the laws of gravity, there must've been a far, far greater amount of mass within the gravity clusters than we could see. It was this prediction of unseen mass that he termed "dunkle Materie" or dark matter [2]. At first, his strange prediction was rejected by many, but once many more observations were taken, it seemed like there truly was an enormous amount of mass in the universe that we could not see or account for.

Thus came the belief in the existence of dark matter- a concept that was created as a placeholder to explain frankly the inconsistencies in the observed effects of gravity. By nature, dark matter is so called "dark" since it does not interact with ordinary matter, nor is it affected by the electromagnetic or strong forces, resulting in it being unperceivable to us with our current level of technology. We cannot see it and we cannot detect it by any currently possible means, but somehow it has been calculated to make up around 80% of all the matter and energy in the universe.

As of now, we can only theorise on the nature of dark matter. An earlier theory proposed the idea of MACHOs- massive astrophysical compact halo objects. These are made of regular, baryonic matter however they emit very little light so are incredibly difficult to detect. A collective term for faint objects like black holes, neutron stars, planets and other very dim stars like some red, white, brown dwarfs. They exist in the halo of galaxies and have been detected through gravitational microlensing. Even so these MACHOs would only account for 20% of the amount of dark matter believed to be present in the universe, with the rest remaining a mystery.



Gravitational lensing, how MACHOs focus light [3]

The current most widely accepted idea is that it is made of a non-baryonic type of particle dubbed WIMPs- weakly interacting massive particles. It is believed that these particles would interact with the weak force as well as gravity and other unknown forces

outside the standard model. They may have even been created thermally in the very early days of the universe like the other particles of the standard model. A hypothetical supersymmetric extension of the standard model happens to have a place for a particle with similar predicted properties, however as of now there is no experimental evidence to support the claim.

Other ideas say that it may be a lighter hypothetical particle named axions, or composed of an entirely different set of lighter particles to the standard model, aptly named the hidden/ dark sector.

Perhaps, the explanation for dark matter exists in a 5th dimension. It could interact with electromagnetism, but us who live in a 3 dimensional world with time being the 4th, would have no way of perceiving it.

Or alternatively, ideas such as Modified Newtonian Dynamics (MOND) challenge the need for dark matter altogether. It introduces fundamental changes to Newton's laws of motion at very low accelerations. Instead of assuming unseen mass to explain the rotation curves of galaxies, MOND suggests that gravity behaves differently on large scales to fit the lines of experimental data without requiring the presence of unseen mass.

In summary, the quest to understand dark matter remains one of the most intriguing challenges in modern astrophysics. While many theories do exist, they are yet to be confirmed experimentally. As observational techniques and particle detection methods continue to advance, we move closer to uncovering the truth behind one of the greatest mysteries of our universe.

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About the Author

Sherlyn Nammi is a King's College London student of Physics with Astrophysics & Cosmology.

Dark Energy, Black Holes, and the End of the Universe: Is there hope for civilisation at the end?

Author: Jason Pettigrew

“Life finds a way”

A surprisingly profound quote from a franchise that revolves around genetically engineered dinosaurs terrorising people in what must be by this point, the world’s most negatively reviewed theme park.

But it begs the question – will life always find a way? It’s certainly made an effort so far. From asteroids, to ice ages, to super volcanic eruptions, life has survived all manner of cataclysm since its birth in Earth’s primordial soup. Yet despite all that, it’s only grown more complex, more intelligent, more diverse – it truly is a marvel. But how long can life keep it up? Is there always a way, or is there a final end to the race?

It is this simple quote that calls me to push this concept to its extreme. Could life survive the ultimate cataclysm – the end of the Universe?

The End (and the Beginning) of the Universe

Before we can answer such a question, we need to know what the end of the Universe looks like. For that, we need to look at its beginning. Approximately 14 billion years ago, the very fabric of space-time, as well as all matter in the universe, was violently ejected from a singularity. This event was the Big Bang. Ever since, our universe has been expanding, with space-time “riding the wave” of this cosmic propulsion.

This expansion was first discovered by Edwin Hubble in 1929, after he observed the red-shifting of light from distant galaxies was greater than that of the light emitted by nearby galaxies. This drew a direct correlation between the distance light has to travel to reach us, and its redshift, implying that space itself is expanding uniformly in all directions. This is likely a bizarre notion to many, but is logical in the context of a Big Bang Universe – again, think of all matter in the universe riding a wave of expanding space-time.

In 1998, astronomers observing distant supernovae found evidence that the universe is in fact expanding at an accelerating rate [1]. This was a surprising revelation given that up until this point, it had been assumed that gravity would be acting against the universe's expansion and that, if there was enough matter present¹, the expansion would eventually stop and even reverse. It is unknown what is causing this acceleration, but according to the law of conservation of energy, you cannot have acceleration without energy. Hence this enigmatic accelerant has been named Dark Energy, due to its unseen nature.

So what does this mean for the end of the Universe?

Well, if dark energy does not decrease, eventually, the universe will expand faster than the speed of light. Galaxies will no longer receive the light of others, and any space faring civilization will be unable to reach other galaxies (assuming that faster than light travel remains an impossibility). The entropy of the universe will increase as dictated by the 2nd law of thermodynamics, and slowly the universe will tend towards a state of disorder, with matter and energy dispersed across the void of space. Galaxies will remain, bound by gravity², but in 100 trillion years, the last stars will die and all that will remain are cold rocks and black holes.

This is the Heat Death of the universe.

Civilisation at the End

It should be noted that discussions of the workings of intelligent life so far into the future are ultimately speculative and frankly, hubristic. To presume that our most fundamental understanding of the universe will meaningfully apply trillions of years into the future, when only 500 years ago we boldly proclaimed that the sun revolved around the Earth - is a notion I find foolish. Nonetheless, I will be so bold as to assume a seemingly fundamental necessity for life: energy.

If life, and by extension a civilisation, in whatever shape it may take, requires energy to survive (as evidenced by everything we understand of our natural world), the Heat Death of the universe seems to spell a pretty bleak end for life as we know it. There are no stars from which to draw energy and any lifeforms of the epoch are confined to the finite

¹ This relates to whether or not the energy density of the universe exceeds the critical density. If it had, the universe would actually begin to contract, resulting in a "Big Crunch".

² An alternative theory poses that if the acceleration of expansion is itself, accelerating, then the Universe will end in a "Big Rip", wherein the expansion of the universe will eventually overcome the nuclear forces, ripping all matter down to only fundamental particles.

resources of the galaxy in which they reside. Eventually, the universe will be littered with dead galaxies that swirl silently around feeding black holes, and from those, nothing will escape.

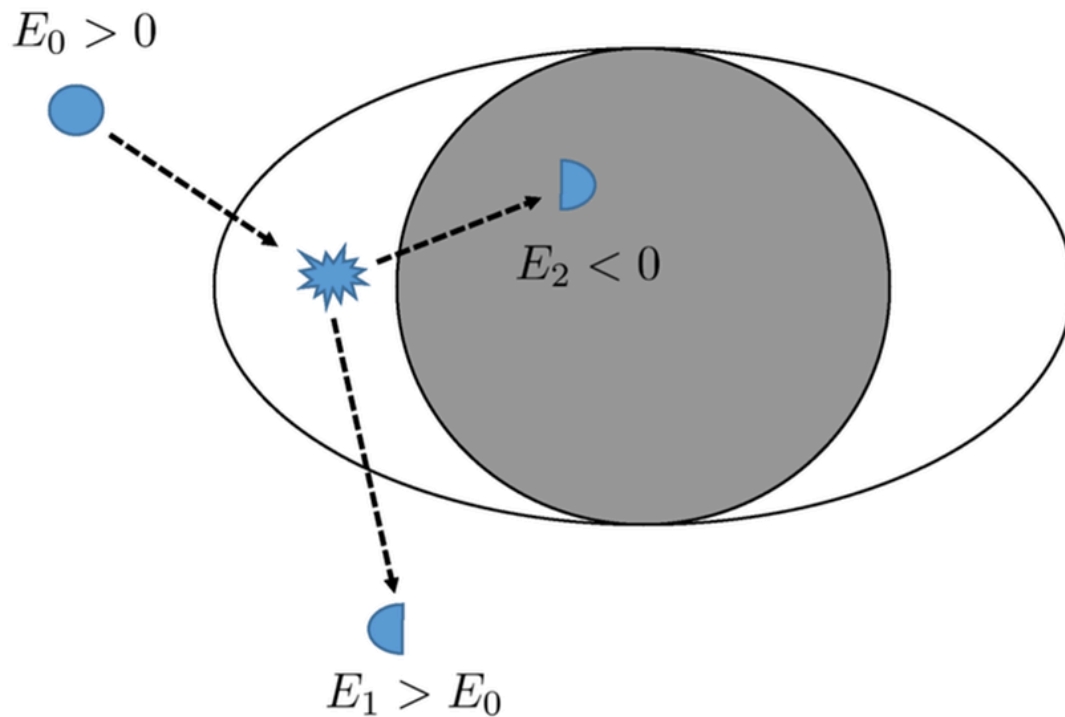
Unless...

The Ironic Light in the Dark: Harvesting Energy from Black Holes

In 1971, English mathematician Roger Penrose proposed a method for extracting the energy of a rotating black hole, now known as the Penrose Process [2]. Many may first object to hearing this - it is frequently repeated, to physicists and non-physicists alike, that nothing can escape the confines of a black hole, not even light, nor energy. The Penrose process does not violate this law however, instead the energy of the black hole itself is extracted.

Many stars in the universe are rotating. If a large rotating star collapses into a black hole, according to the conservation of momentum, the angular momentum of the star must be conserved and thus it is hypothesised that the black hole created from the star's collapse would also be rotating. This creates a volume around the black hole, beyond the event horizon called the ergosphere, in which space-time is dragged along with the immense gravity of the black hole (a phenomena named frame dragging). These are known as Kerr black holes.

Penrose proposed that if an object was fired into the ergosphere and broke into two, with one piece entering the black hole and one exiting, then due to the conservation of momentum, the black hole would transfer energy to the exiting piece, at the cost of its own rotational energy. A simplistic diagram is presented below [3].



A simplistic diagram illustrating Penrose's theory.

A sufficiently advanced future civilisation would likely be able to convert the kinetic energy of the escaping object into electrical energy.

Another potential method of energy extraction is the Blandford–Znajek process. This utilises the fact that the extreme pressure and friction between matter in the accretion disk of a Kerr black hole, ionises the matter circulating due to the extreme heat generated. When charge moves, magnetic fields are generated. Hence, magnetic field lines run along the poloidal axis of the black hole and are twisted by frame dragging in the ergosphere. This accelerates the charged particles to incredibly high speeds, ejecting them from the poles of the black hole and causing them to emit high energy gamma rays. This is believed to be the mechanism behind quasars.

A civilisation capable of efficiently harnessing these high energy jets could theoretically produce more energy in a second than the globe consumes in a year [4].

Conclusion

So, is this it? The solution to life's survival? Using black holes, could life sustain itself indefinitely? Sadly, not quite. Both the Penrose process and the Blanford-Znajek process fall prey to that timeless pest – the conservation of energy. If you're extracting energy, it must be coming from somewhere. In this case, both methods extract the rotational energy of the black hole itself, meaning that eventually, a civilisation using either of these methods would find their black hole had stopped rotating, leaving it devoid of usable energy.

However, even if this was not the case, due to Hawking radiation, even black holes will eventually dissipate (albeit after an inconceivably long time). Nonetheless, these methods could allow life to survive for a long time beyond the death of the last stars.

So, will life always find a way? For now, the answer is a tentative no, however given trillions of years, who knows what might happen. Perhaps dark energy will begin to decrease as some physicists have suggested, or perhaps an advanced species, human or otherwise, will discover some fundamental secret of the universe that allows for life's indefinite survival.

In any case, the end of the universe is a long way off, and we have many more pressing problems to focus on if we have any hope of getting that far.

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About the Author

Jason Pettigrew is a Physics student completing his final year of Bachelor's at King's College London. He is interested in science fiction novels and nuclear physics, and fascinated by the cosmology of the universe.

RMP 2024 Reflections

Author: Jana Bazeed

In 2024, the Global Society of Young Physicists (GSYP) hosted its flagship Research Mentorship Programme (RMP). We reached out to some of its alumna to reflect about their experiences.

Designed to provide aspiring physicists with a substantive research experience, the programme facilitated mentorship by pairing high-school students with university students, allowing them to explore advanced topics in various fields of physics.

A defining feature of RMP is its emphasis on mentorship and collaboration. Participants engaged in small-group online lectures led by physics students, where they explored complex subjects such as quantum mechanics and mathematical physics. Beyond attending lectures, students actively contributed to research projects, authored academic papers, and presented their findings to their peers, thereby gaining invaluable experience and academic confidence.

Alumna Elleanore Phillips found the Mathematical Physics stream particularly enriching:

"Participating in the Mathematical Physics GSYP stream was an exceptionally rewarding and stimulating experience. I would highly recommend it to anyone interested in the intersection of physics and mathematics. The programme involved small-group lectures with an outstanding mentor, followed by collaborative research on a chosen topic, culminating in the writing of a research paper and a presentation to other groups. The supportive environment significantly enhanced my learning experience, and I am incredibly grateful for it. In fact, my experience was so positive that I have applied to return for the research internship this summer."

For Beatriz Peneda, another alumna, the access to quality mentorship was key:

"During the course, I was fortunate to learn from Arya. Her profound understanding of quantum mechanics enabled her to explain intricate concepts in a way that made them seem intuitive. This was an entirely new experience for me, especially given the highly non-intuitive nature of quantum mechanics. The foundational knowledge I gained has since allowed me to engage with university-level textbooks on the subject with comprehension and confidence."

Beyond the academic rigor of the programme, participants also appreciated the supportive and inclusive environment, which fostered curiosity and intellectual engagement. Chiara Balestrini reflected:

"Looking back, RMP 2024 enabled me to step outside my comfort zone and deepen my understanding of mathematics and physics."

Balestrini also advised prospective mentees to ask questions and stay curious:

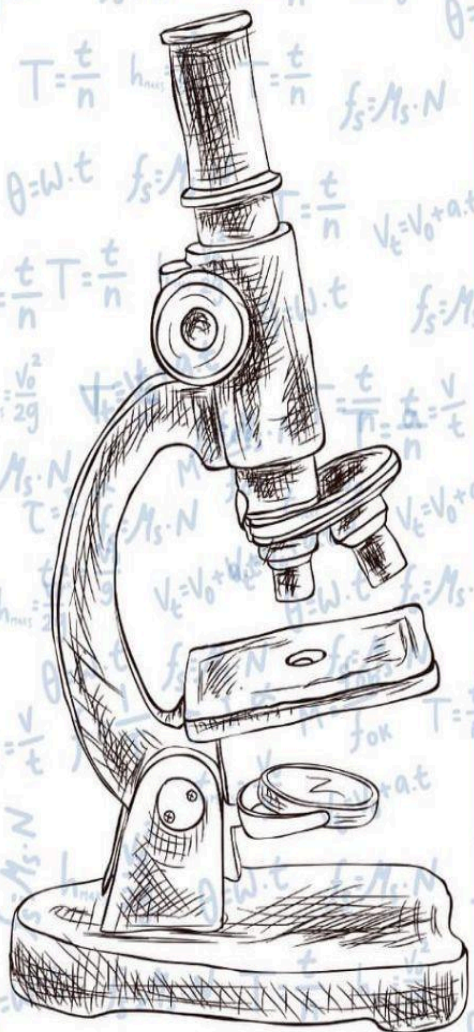
"My advice to future applicants is to never hesitate to ask questions—whether out of curiosity or uncertainty. Inquiry is the key to deeper exploration and understanding. As Einstein himself stated: 'The important thing is not to stop questioning. Curiosity has its own reason for existing.'"

With overwhelmingly positive feedback from participants, the GSYF Research Mentorship Programme has proven to be an invaluable opportunity for aspiring physicists. By engaging in substantive research, working with dedicated mentors, and collaborating with like-minded peers, students have gained significant academic and professional experience.

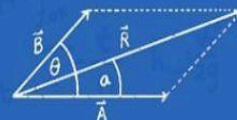
About the Author

Jana Bazeed is a final year MSci Physics and Philosophy student at King's College London and award-winning journalist with a keen interest in science communication, and the intersection of science, philosophy, and society.

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RMP 2025



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$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

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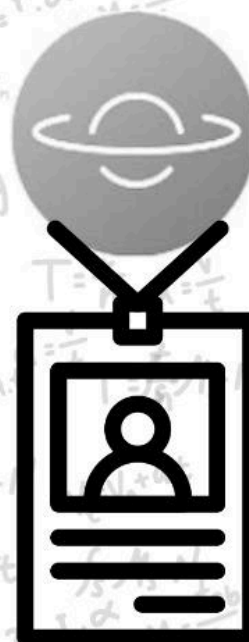
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CONTRIBUTIONS

Arya Lal Gonullu, Chief Executive of GSYF, named the science magazine after Galileo Galilei's groundbreaking work, *Sidereus Nuncius* (The Starry Messenger). In this historic book, Galileo had unveiled three revolutionary discoveries he made with it: Sunspots, the Moons of Jupiter and that the Moon is not a perfect sphere. Gonullu hopes that physics will soon experience a new “Starry Messenger” breakthrough through the new rising stars of the field.

Editor of The Starry Messenger: Sherlyn Nammi

Deputy Editors of The Starry Messenger: Leah Levy, Jana Bazeed, Piyamini Tenahandi

Poster Design : Chiara Allegri, the Head of Marketing at GSYF

Interested in writing for the next issue of *The Starry Messenger*? Send your insightful research, or creative take on the latest in physics to arya.gsyp@gmail.com and be a part of our growing community!

Deadline: Submissions close on November 15 , 2025 for the next issue.



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